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THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the reduction and the limit computation are both inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Limit of a directed system of schemes with affine transition morphisms and quasi-compact quasi-separated structure morphisms. Same chapter, the set-up of the limit X of the X_i over S and the pullbacks of the sheaves F_i to X. Supplies the objects appearing in the asserted identification.
- Cohomology of a limit of schemes and the description of higher direct images as sheafified presheaves. Same chapter, the commutation of cohomology with the limit and the description of the higher direct images by their values on the objects of the base. Supplies the two ingredients of the computation.

Source reference. Original lemma 03Q8: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7510–7510. Navigation pointer invoked at proof line 7550; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 09YQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7400–7400. Navigation pointer invoked at proof line 7480; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $X = \lim_{i \in I} X_i$ be a limit of a directed system of schemes over S with affine transition morphisms $f_{i'i} : X_{i'} \to X_i$. We assume the structure morphisms $g_i : X_i \to S$ and $g : X \to S$ are quasi-compact and quasi-separated. Let $(\mathcal{F}_i, \varphi_{i'i})$ be a system of abelian sheaves on $(X_i, f_{i'i})$. Denote $f_i : X \to X_i$ the projection and set $\mathcal{F} = \operatorname{colim} f_i^{-1}\mathcal{F}_i$. Then*

$$\operatorname{colim}_{i \in I} R^p g_{i,*}\mathcal{F}_i = R^p g_*\mathcal{F}$$

for all $p \geq 0$.

Proof. Recall (Lemma 03Q8) that $R^p g_{i,*}\mathcal{F}_i$ is the sheaf associated to the presheaf $U \mapsto H^p_{étale}(U \times_S X_i, \mathcal{F}_i)$ and similarly for $R^p g_*\mathcal{F}$. Moreover, the colimit of a system of sheaves is the sheafification of the colimit on the level of presheaves. Note that every object of $S_{étale}$ has a covering by quasi-compact and quasi-separated objects (e.g., affine schemes). Moreover, if U is a quasi-compact and quasi-separated object, then we have

$$\operatorname{colim} H^p_{étale}(U \times_S X_i, \mathcal{F}_i) = H^p_{étale}(U \times_S X, \mathcal{F})$$

by Theorem 09YQ. Thus the lemma follows. □